

Performance
Testing

Date	12 July 2026
TeamID	SWTID-2026-9030
ProjectName	Credit Card Approval Prediction
MaximumMarks	5 Marks

PerformanceTesting
Performance testing evaluates howthe system behaves under expected andpeak load conditions. Complete the tests below to check theapplication'sresponsetime, resourceusage, andoverallstability.

Step 1: Testing Overview

Fiel d	Detail s
Testing ToolUsed	Postman, Swagger UI
Type ofTesting	Functional& Performance Testing
Target Modules / API	Data Preprocessing, Model Prediction, Result Generation API
Test Environment	Local FastAPI + Streamlit
Test Duration	10 Minutes

Step 2: Test Scenarios

S. N o	TestScenario Description	No. of Virtual Users	Duration(sec)	Expected Outcome
1	Uploadapplicantdatafor prediction	1	5	Datauploadedand processedsuccessfully
2	Concurrent predictionrequests	5	8	Predictions returned without failure
3	Model inference and result check	1	5	Modelreturnsvalid approval/rejection result

Step 3: Performance Test Results

S. N	Met	Target	Actual	Status (Pass/Fai	Rema
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o	ric	Value	Value	l)	rks
1	Response Time (Avg)	< 2 seconds	1.2 sec	Pass	Within limits
2	Response Time (Max)	< 3 seconds	2.8 sec	Pass	Slightly high under load
3	Throughput (Requests/sec)	> 10 req/sec	12 req/sec	Pass	Stable performance
4	Error Rate	< 1%	0%	Pass	No errors observed
5	CPU Utilization	< 70%	55%	Pass	Normal usage
6	Memory Utilization	< 70%	60%	Pass	Stable memory usage

Step 4: Observations & Analysis

Key Findings:

- All core functions (data upload, prediction, result generation) responded within acceptable time limits.
- System handled concurrent requests without errors or crashes.

Bottlenecks Identified:

- Slight increase in response time observed during concurrent prediction requests.

Optimization Steps Taken:

- Optimized API request handling to reduce processing overhead.
- Improved model loading to reduce inference latency under load.

Step 5: Screenshots / Evidence

[Insert screenshots of your Postman/Swagger UI test runs, response times, and terminal logs here.]